

Modeling Computation

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Preface

These exercises were assembled to accompany a Discrete Mathematics course at Calvin College.

1 Alphabets, Languages, and Grammars

1.1 Terminology

- An **alphabet** (or **vocabulary**) is just a finite, non-empty set. It's elements are called **letters** or **symbols**.
- A **word** (or **string**) is a finite sequence of symbols.
- A word consisting of no symbols is called the **empty word** and denoted λ . (Think "".)
- The set of all words using exactly n symbols from V (repetition allowed) is denoted V^n .
- The set of all words using symbols in alphabet V is denoted V^* . (So $V^* = V^0 \cup V^1 \cup V^2 \dots$)
- A **language** over V is a subset of V^* .

Exercises

1. Let $V = \{0, 1\}$. What is V^0 ? What is V^2 ? What is V^* ?
2. Let $V = \{2\}$. What is V^0 ? What is V^2 ? What is V^* ?
3. Can $\emptyset$ (empty set) be a letter? an alphabet? a word? a language?

1.2 (Phrase-Structure) Grammars

A phrase-structure grammar consists of

- **alphabet:** V
- **start symbol:** $S \in V$
- **terminal symbols:** $T \subset V$
 - $N = V - T$ is the set of **nonterminal symbols**
- **production rules:** $P \subseteq (V^* - N^*) \times V^*$
 - usually write elements of P as $u \rightarrow v$ rather than (u, v) .
 - $V^* - N^*$ says that u must contain at least one nonterminal.
 - v can be any combination of terminals and nonterminals (including the empty string).

Examples

Note: in each of these examples, capital letters are used for nonterminals and lower case letters or digits for terminals. That makes it easy to remember, but it is not a requirement of the definition of a grammar.

Grammar 1 (G_1): alphabet: $\{a, b, A, B, S\}$, terminals: $\{a, b\}$, start symbol: S , production rules:

- $S \rightarrow ABa$
- $A \rightarrow BB$
- $B \rightarrow ab$
- $AB \rightarrow b$

Grammar 2 (G_2): alphabet: $\{S, A, a, b\}$, terminals: $\{a, b\}$, start symbol: S , production rules:

- $S \rightarrow aA$
- $S \rightarrow b$
- $A \rightarrow aa$

Grammar 3 (G_3): alphabet: $\{S, 0, 1\}$, terminals: $\{0, 1\}$, start symbol: S , production rules:

- $S \rightarrow 11S$
- $S \rightarrow 0$

Grammar 4 (G_4): alphabet: $\{S, A, B, C, a, b, c\}$, terminals: $\{a, b, c\}$, start symbol: S , production rules:

- $S \rightarrow AB$
- $A \rightarrow Ca$
- $B \rightarrow Ba$
- $B \rightarrow Cb$
- $B \rightarrow b$
- $C \rightarrow cb$
- $C \rightarrow b$

1.2.1 Derivations and Languages

The rules of a grammar are used to derive strings of terminals (elements of T^*) as follows.

- If $w_0 \rightarrow w_1$ is a rule, then $lw_0r \Rightarrow lw_1r$ for any strings l and r .
- $a \xRightarrow{*} b$ is defined recursively. $a \xRightarrow{*} b$ if either
 - $a \Rightarrow b$, or
 - $\exists c (a \Rightarrow c \wedge c \xRightarrow{*} b)$

Basically the production rules are “rewrite rules” that allow us to replace the left side of the rule with the right side. A derivation is a sequence of rewrites.

The language of a grammar G is denoted $L(G)$ and contains all strings of terminals that can be derived from the start symbol:

$$L(G) = \{w \in T^* \mid S \xRightarrow{*} w\}$$

1.2.1.1 Exercises

4. Show that using Grammar 1, $ABa \xRightarrow{*} abababa$.
5. Is $abababa \in L(G_1)$, the language generated by Grammar 1?
6. What is $L(G_2)$?
7. What is $L(G_3)$?
8. Generate several words using G_4 .
9. Create a grammar G_5 such that $L(G_5) = \{0^n 1^n \mid n = 0, 1, 2, \dots\}$
10. Create a grammar G_6 such that $L(G_6) = \{0^n 1^n \mid n \in \mathbb{Z}^+\}$
11. Create a grammar G_7 such that $L(G_7) = \{0^m 1^n \mid m, n \in \mathbb{N}\}$